

pktPCAP — IT Administrator Guide

[Packet Capture Analysis Platform](#) · [Administrator Reference](#)

Overview

This guide covers the full lifecycle of operating pktPCAP as an administrator: system requirements, installation, SSL configuration, authentication setup (local and SAML SSO), user and role management, AI provider configuration, logging, API reference, and troubleshooting.

Administrator access is required for all configuration tasks described in this document. The Settings panel (`/settings`) is restricted to Admin-role users only.

System Requirements

Component	Requirement
Operating System	Linux (Amazon Linux 2, RHEL 8+, Ubuntu 20.04+)
Python	3.8 or later
Network	HTTPS port (default 8765) accessible from analyst workstations
Disk	Minimum 20 GB free — capture files can be large
RAM	4 GB recommended (large capture parsing is memory-intensive)
Outbound Internet	Required for AI API calls (OpenAI / Anthropic)
Capture Tools	tshark on remote hosts for live feed capture (optional)
SSL Certificate	Valid certificate/key pair — wildcard or host-specific

Installation

1. Clone the Repository

```
git clone git@github.com:<your-org>/pktpcap.git /opt/pktpcap
```

2. Create a Python Virtual Environment

```
cd /opt/pktpcap
python3 -m venv venv
./venv/bin/python -m pip install -r service/requirements.txt
```

Important: Always install packages using `./venv/bin/python -m pip install` rather than the `pip` binary directly. If multiple Python environments are present on the host, the `pip` shebang may point to the wrong interpreter.

3. Place SSL Certificates

```
mkdir -p /opt/pktpcap/ssl
cp /path/to/server.crt /opt/pktpcap/ssl/server.crt
cp /path/to/server.key /opt/pktpcap/ssl/server.key
chmod 600 /opt/pktpcap/ssl/server.key
```

SSL is auto-detected on startup based on file presence alone — no configuration toggle is required.

4. Initialize the Database

Start the server once manually to initialize the SQLite database and generate the secret key:

```
cd /opt/pktpcap/service  
../venv/bin/python server.py
```

Confirm it starts cleanly, stop it (`Ctrl+C`), then configure systemd.

systemd Service

pktPCAP runs as a managed systemd service for automatic start on boot and supervised restart on failure.

Service Unit File

Create `/etc/systemd/system/pktpcap.service` :

```
[Unit]
Description=pktPCAP Packet Capture Analysis Platform
After=network.target

[Service]
Type=simple
User=your-service-user
WorkingDirectory=/opt/pktpcap/service
ExecStart=/opt/pktpcap/venv/bin/python server.py
Restart=on-failure
RestartSec=5

[Install]
WantedBy=multi-user.target
```

Enable and Start

```
sudo systemctl daemon-reload
sudo systemctl enable pktpcap
sudo systemctl start pktpcap
sudo systemctl status pktpcap
```

Service Commands

Action	Command
Start	sudo systemctl start pktpcap
Stop	sudo systemctl stop pktpcap
Restart	sudo systemctl restart pktpcap
Status	sudo systemctl status pktpcap
Live logs	sudo journalctl -u pktpcap -f

Tip: Administrators can also restart the service from the pktPCAP Settings panel under **Service Controls**, without SSH access.

SSL / TLS Configuration

pktPCAP enforces HTTPS. TLS is activated automatically when certificate files are detected — there is no database flag or environment variable.

Certificate Paths

File	Path	Description
Certificate	ssl/server.crt	PEM-format certificate or chain
Private Key	ssl/server.key	PEM-format private key

Paths are relative to the repository root. If either file is missing on startup, the server falls back to HTTP — avoid this in production.

Converting a PFX/P12 Certificate

```
# Extract the certificate
openssl pkcs12 -in your-cert.pfx -nokeys -out ssl/server.crt

# Extract the private key
openssl pkcs12 -in your-cert.pfx -nocerts -nodes -out ssl/server.key
```

Security: Protect the private key. Set permissions to `600`, ensure it is owned by the service user, and never commit it to version control.

Certificate Renewal

Replace the files in place and restart the service:

```
sudo systemctl restart pktpcap
```

Authentication

pktPCAP supports two authentication methods simultaneously:

- **Local Authentication** — username/password stored in SQLite (bcrypt hashed), managed from the Settings panel.
- **SAML 2.0 SSO** — enterprise identity provider integration (see next section).

Authentication state is maintained via server-side Flask sessions. All sessions are invalidated on logout or service restart.

SAML SSO Configuration (Okta)

pktPCAP integrates with any SAML 2.0 identity provider using the `python3-saml` library. These steps document configuration for Okta, but the process is similar for other IdPs.

Prerequisite: `python3-saml` must be installed in the pktPCAP virtualenv: `./venv/bin/python -m pip install python3-saml`

Step 1 — Create an Okta App Integration

1. In Okta Admin: **Applications** → **Create App Integration** → **SAML 2.0**
2. Set **Single Sign-On URL** to: `https://<your-server>:<port>/api/auth/saml/callback`
3. Set **Audience URI (SP Entity ID)** to: `https://<your-server>:<port>/api/auth/saml/metadata`
4. Optionally add an **Attribute Statement**: name `role`, mapped to `appuser.Role` (enables role sync)

Step 2 — Configure pktPCAP Settings

In Settings → SSO / SAML Settings:

Field	Value
IdP Entity ID	From Okta app → Sign On → SAML Setup (Issuer)
IdP SSO URL	Identity Provider Single Sign-On URL from Okta
IdP Certificate	X.509 certificate downloaded from Okta app settings
SP Entity ID	<code>https://<your-server>:<port>/api/auth/saml/metadata</code>

First Login Behavior

SSO users are provisioned automatically on first login. If Okta sends a `role` attribute, it is applied. If no role attribute is present, the user receives the **Admin** role by default — adjust via Settings → Users if needed.

Role Sync

On every subsequent SSO login, pktPCAP reads the `role` attribute from Okta and syncs it to the local user record. Role changes in Okta propagate to pktPCAP on next login — no manual update required.

Service Provider Metadata

SP metadata XML is available at: `/api/auth/saml/metadata` — provide this URL to your IdP during app setup if it supports metadata import.

User Management

User accounts are managed from **Settings** → **Users** (Admin only).

Creating a User

- 1. Navigate to Settings → Users → **Add User**
- 2. Enter username, email address, and initial password
- 3. Assign a role: Admin, Analyst, or Viewer
- 4. Set status to **Active** and save

Editing, Deactivating, and Password Reset

- **Edit** any user row to change username, email, role, or status
- **Reset Password** — set a new password for local-auth users (SSO users do not use passwords)
- **Deactivate** — set status to Inactive to block login without deleting the account

Role Reference

Permission	Admin	Analyst	Viewer
Upload PCAP files	✓	✓	X
Connect live feeds	✓	✓	X
Run AI analysis	✓	✓	Read-only
View analysis data	✓	✓	✓
Export results	✓	✓	X
Take screenshots	✓	✓	X
Access Settings panel	✓	X	X
Manage users	✓	X	X
Configure AI provider	✓	X	X
Configure SSO	✓	X	X
View logs	✓	X	X
Restart service	✓	X	X

All role enforcement is **server-side**. UI elements are hidden from non-Admin users as a convenience, but every API endpoint independently enforces role requirements.

AI Provider Configuration

Configure your preferred AI backend from **Settings** → **AI Configuration**.

OpenAI

Provide your OpenAI API key and select the model (e.g., `gpt-4o` , `gpt-4-turbo`). The key is stored encrypted in the database and never exposed to the browser.

Anthropic Claude

Provide your Anthropic API key and select the model (e.g., `claude-sonnet-4-5`). Configuration is identical to OpenAI.

Privacy

Only structured packet metadata and analyst queries are sent to the AI provider. Raw packet bytes, payload contents, and capture files are never transmitted — all parsing happens locally on the server.

Testing the Connection

Use **Test Connection** in Settings → AI Configuration to send a probe request and confirm the API key and model are valid.

Logs & Monitoring

Application logs are viewable in **Settings** → **Logs**.

Log Levels

Level	What is Logged
DEBUG	Full request/response detail — verbose, for troubleshooting only
INFO	Normal operation: logins, uploads, analysis completions (recommended)
WARNING	Recoverable issues: auth failures, partial parse results
ERROR	Non-fatal errors requiring attention
CRITICAL	Service-threatening failures

Change the log level in real time from Settings — the change takes effect immediately without a service restart.

API Endpoint Reference

All endpoints require a valid authenticated session. Endpoints marked **[Admin]** return `403 Forbidden` for non-Admin sessions.

Authentication

Method	Endpoint	Description
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POST	/api/login	Local username/password login
POST	/api/logout	Invalidate current session
GET	/api/auth/current-user	Returns current user info and role
GET	/api/auth/saml/login	Initiates SAML SSO redirect
POST	/api/auth/saml/callback	Receives SAML assertion from IdP
GET	/api/auth/saml/metadata	SP metadata XML

User Management [Admin]

Method	Endpoint	Description
GET	/api/users	List all users
POST	/api/users	Create a new user
PUT	/api/users/<id>	Update user (role, status, email)
DELETE	/api/users/<id>	Delete user
POST	/api/users/<id>/reset-password	Reset user password

Settings & Configuration [Admin]

Method	Endpoint	Description
GET	/api/settings	Retrieve current settings
POST	/api/settings	Update settings
GET	/api/logs	Retrieve log entries
DELETE	/api/logs	Clear all logs
POST	/api/logs/level	Change runtime log level
POST	/api/restart	Restart the server process

Analysis & Capture

Method	Endpoint	Description
POST	/api/ai	Submit a query to the AI engine
POST	/api/feed/<name>	Receive live packet feed stream
GET	/api/feeds	List active feed sessions
POST	/api/save-image	Save a dashboard image server-side
POST	/api/screenshot	Capture a full dashboard screenshot

Troubleshooting

Service Won't Start

Check the journal: `sudo journalctl -u pktpcap -n 50 --no-pager`

Common causes:

- **Missing Python module** — run `./venv/bin/python -m pip install -r service/requirements.txt`
- **Port already in use** — check with `ss -tlnp | grep 8765`
- **SSL cert missing or malformed** — confirm files exist at `ssl/server.crt` and `ssl/server.key`

SAML SSO Login Fails with "Invalid Response"

- Verify the SP Entity ID in your IdP matches exactly (including port number and trailing path)
- Confirm the IdP certificate in pktPCAP Settings matches the current signing cert in your IdP
- Check that the server's clock is synchronized — SAML assertions have a short validity window and clock skew causes validation failures (`chronyc tracking` or `timedatectl`)

AI Analysis Returns an Error

- Verify the API key is valid — use **Test Connection** in Settings
- Confirm the server has outbound internet access to the AI provider endpoint
- Check the log viewer in Settings for the specific error message from the AI API

Live Feed Not Appearing After Running tshark Command

- Confirm the feed token in the command is current — regenerate from the Live Feeds page if needed
- Verify the remote host can reach the pktPCAP server on the configured port
- Check that `tshark` is installed on the remote host: `tshark --version`
- Confirm the capture interface name is correct for the remote host

Large File Uploads Time Out

If very large captures fail to upload, the issue is typically a client-side or reverse-proxy timeout. If pktPCAP is behind a load balancer or reverse proxy, increase the upstream timeout. Consider pre-filtering with Wireshark or tshark to reduce file size before uploading.

"SAML Not Available" in Settings

The `python3-saml` library is not installed in the virtualenv. Install it:

```
./venv/bin/python -m pip install python3-saml
sudo systemctl restart pktpcap
```